

FEC Data Exploration

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→	Overview	
→	Data Overview	
→	Alignment Tool	
→	Data Analysis with Alignment Tool	
→	Classifiers	



FEC Data Overview

The Federal Election Commission (FEC) has bulk datasets on a variety of campaign finance topics such as individual contributions and campaign spending.

I selected a dataset which contained information on political action committee's (PACs) contributions to candidates and independent expenditures.

The dataset contained information like:

- Committee ID
- Candidate ID
- Transaction amount
- Transaction Type
 - Can be used to find if money was to support or oppose
- Memo Text/ Recipient Name

*Additionally I used FEC dataset to connect metadata to committees and candidates (i.e. name, party, state)

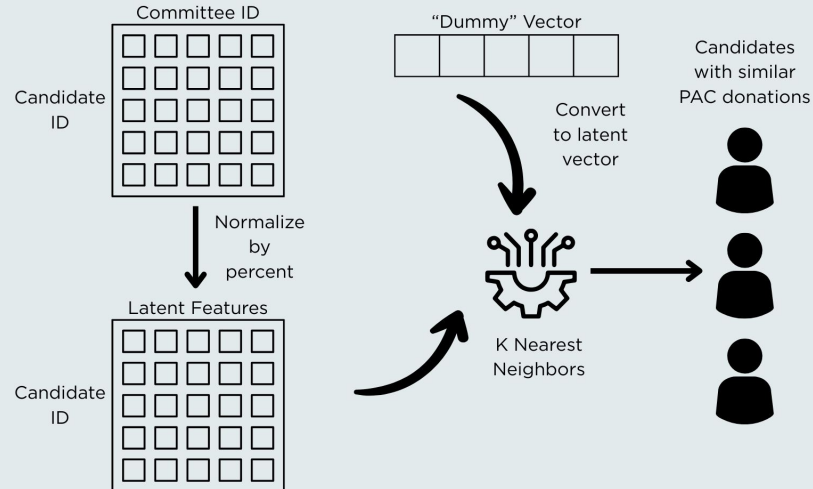
Interaction Matrix/Factorization

- Constructed interaction matrices for candidates and committees by total amount given in a cycle
 - Normalized
- Factorized using Truncated SVD

Alignment Tracking

- Designed to extract signal from factorized interaction matrices
- Build 'dummy' vectors of hypothetical candidates or committees based on contributions
- Find nearest neighbors to 'dummy vector'

These neighbors to these hypothetical candidates/committees can be used track alignment with certain interest and map the latent space.



→ Alignment Tool Examples

Example: Environmentalists



John Sarbanes
MD - 3, Dem



Cory Bush
MO - 7, Dem



**Alexandria
Ocasio Cortez**
NY - 14, Dem



Ayanna Pressley
MA - 7, Dem

**Aligned
Committees:**



→ Alignment Tool Analysis



-Indicates Potential Committee Assignment Conflicts

Coal/Oil Barons



Chris Stewart
UT - 2, Rep

WAH
OH

*House subcommittee on
interior, Environment,
and Related Agencies*



Jay Obernolte
CA - 23, Rep

WAH
OH

*House Committee on
Energy and Commerce*



Mike Johnson
Speaker of the House
LA - 4, Rep

WAH
OH

Speaker of the House



Michael McCaul
CA - 23, Rep

Aligned Committees:



Retail Warriors



Kweisi Mfume
MD - 7, Dem

WAH
OH

*House committee on
education and workforce
House committee on Small
Business, Vice Chair*



Stacey Plaskett
GA - 12, Dem



Andrew Clyde
GA - 9, Rep

WAH
OH

*House subcommittee on
labor health and human
services education and
related agencies*



Rick Allen
GA - 12, Rep

WAH
OH

*House committee on
education and the
workforce, chairman
House committee on
energy and commerce*

Aligned Committees:



→ Classifiers

Classifiers

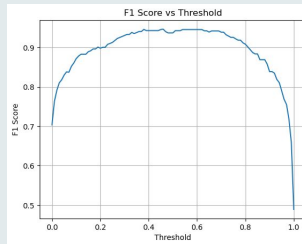
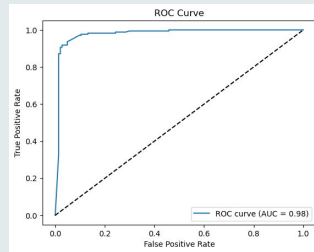
These ultimately help to uncover what signal can be uncovered by this transaction data.

Is Republican?

```
=== Random Forest Model ===
Accuracy: 0.9333333333333333
```

	precision	recall	f1-score	support
False	0.91	0.95	0.93	144
True	0.96	0.92	0.94	171
accuracy			0.93	315
macro avg	0.93	0.93	0.93	315
weighted avg	0.93	0.93	0.93	315

Number of test samples: 315

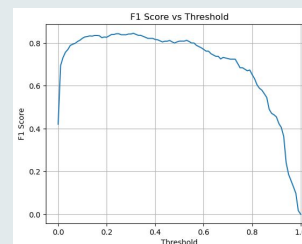
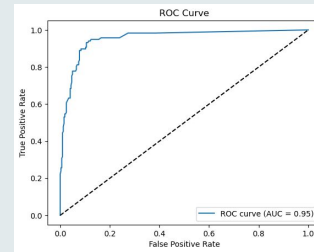


Will Win?

```
=== Random Forest Model ===
Accuracy: 0.9028501138952164
```

	precision	recall	f1-score	support
False	0.92	0.95	0.93	322
True	0.84	0.78	0.81	117
accuracy			0.90	439
macro avg	0.88	0.86	0.87	439
weighted avg	0.90	0.90	0.90	439

Number of test samples: 439

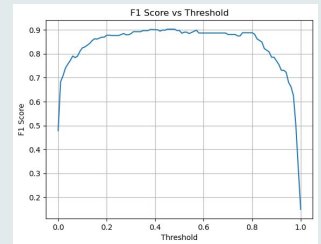
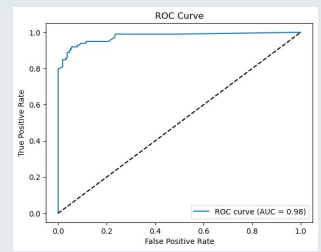


Is Incumbent?

```
=== Random Forest Model ===
Accuracy: 0.9365079365079365
```

	precision	recall	f1-score	support
False	0.95	0.96	0.95	216
True	0.92	0.88	0.90	99
accuracy			0.94	315
macro avg	0.93	0.92	0.93	315
weighted avg	0.94	0.94	0.94	315

Number of test samples: 315



→ Thank you

Any Questions?